

Thirulok Sundar Mohan Rasu

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Education

University of Michigan — Ann Arbor

M.S., Electrical and Computer Engineering.

Relevant course: *Matrix Methods for Signal Processing and Machine Learning*.

Aug 2025 – May 2027

GPA: 4.0 / 4.0

Indian Institute of Technology (ISM) Dhanbad

B.Tech., Electrical Engineering. Member: *CyberLabs (Machine Learning)*

Dec 2021 – May 2025

GPA: 8.30 / 10

Skills

Languages: Python, C++, C, MATLAB.

Areas: Embedded Programming, Hardware Integration, Testing & Debugging, Machine learning, Computer Vision, 3D/4D Reconstruction, LLMs, Robotics, Signal Processing

Tools & Frameworks: PyTorch, TensorFlow, Linux, Git/GitHub, OpenCV, Arduino, Simulink, AWS-S3, ROS2

Experience

University of Michigan, Ann Arbor

Research Assistant

Aug 2025 – Present

- Creating a 4D dataset for robot perception and learning; designing data collection pipelines and evaluation protocols.
- Working on 3D depth estimation algorithms and collecting data for model training with robot arm and tactile sensor.

Carnegie Mellon University

Research Intern (Dr. Arun Balajee Vasudevan)

Jan 2024 – Sep 2025

- Developed a benchmark audio dataset using multiple TTS models with paired real/fake audios; engineered pipelines for generation, labeling, and evaluation. Paper submitted to **Interspeech 2026 Conference** (under review).
- Trained a foundational audio-language model (CLAP) and used transcript-based detection (Llama-8b) to identify synthetic audio
- Visualized audio waveforms and experimented with background noise removal/ speech enhancement techniques.

University of Central Florida

Research Intern (Prof. Yogesh Rawat)

Feb 2024 – Dec 2024

- Created an annotated video-activity dataset and fine-grained evaluation metrics for actor/action/object/environment classes for LLM fine-tuning.
- Designed targeted question types and evaluation methods for robust activity understanding.

Mowito

Robotics Software Intern

Mar 2023 – Jul 2023

- Implemented perceptual-hashing to deduplicate images in time-sorted training datasets, improving model training efficiency.
- Built a Dataset class supporting multiple annotation types (instance, picking, crate) and visualization tools; integrated into production pipelines.
- Implemented order-completion time estimation and Leiden clustering for crate grouping to optimize order fulfillment.

Projects

SO-101 Robot Arm Assembly & Imitation Learning

Assembled and integrated a 3D-printed SO-101 robot arm; collected teleoperation dataset for manipulation tasks; trained Action Chunking Transformer (ACT) policy via imitation learning and evaluated on real hardware.

Implementation of MPPT control techniques (MATLAB & Arduino)

Final-year thesis: Implemented Perturb & Observe and Incremental Conductance MPPT for solar panels; real-time embedded control via Arduino with MATLAB simulation, system verification, and debugging.

Challenges in Multi-view 3D Scene Reconstruction

Analyzed failure cases of MAST3R for multi-view reconstruction under challenging capture conditions; collected a small scene dataset and documented limitations.

Image Classification using Few-Shot Graph Neural Networks

Optimized few-shot GNN pipelines for classification; created dataset of fruit diseases, designed node features and adjacency extraction using deep models.

Bar-Code Detection using OpenCV

Classic CV pipeline to detect barcodes on grocery items and classify bounding boxes by barcode readability status.